

Towards sustainable agriculture: A blueprint for European KPI-based farm-level assessment

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ABSTRACT

Key Performance Indicator (KPI) methodologies have emerged as potent tools for estimating sustainability performance in agriculture. With the proliferation of data availability, there is a growing opportunity to conduct sustainability assessments at the farm level and transition from traditional practice-based policies to performance-based policies. In this paper, we propose a blueprint for a Europe-wide KPI-based farm level sustainability performance assessment and monitoring system. By drawing insights from literature on farm-level sustainability assessment systems and tools, we synthesized eight guiding principles and a 5-step process framework for developing and operationalizing a European Agricultural KPI system built around the potential advantages of performance-based policies. The guiding principles include defining sustainability, involving stakeholders, ensuring applicability across sectors and countries, using normative benchmarking, and implementing decision support approaches. The proposed process framework can guide public and private stakeholders in Europe through the establishment of sustainability goals, modeling linkages between objectives and available data, KPI development, online user interface creation, and adaptive management. By leveraging advanced analytics and the integration of diverse agricultural data sources, our system aims to provide policy-makers with the opportunity to instrumentalize performance-based financial support mechanisms to incentivize the adoption of sustainable agricultural practices among farmers. The essential components of a policy structure supporting the effective and ethical implementation of the proposed KPI system are discussed.

1. Introduction

Globally, efforts to enhance food production have aimed to reduce hunger and promote economic development (Byerlee et al., 2009; Wiebe et al., 2020), leading in recent decades to agricultural policies that promote intensification, specialization, modernization, and increased scale, particularly in the global north (de Roest et al., 2018). Food production has grown at an average annual rate of about 2.2% during the past 60 years (Fuglie, 2018), though with huge variations across the world (Tuninetti et al., 2020). However, this increase in food production had wide-ranging social, economic, and environmental impacts (IPBES, 2019; Kanianska, 2016). Intensification of agriculture has led to substantial losses of fertile soil due to erosion (Montanarella et al., 2016) and decreased soil quality (Kopittke et al., 2019; Tsiafouli et al., 2014). Transformation of natural habitats to agricultural land has lowered

habitat quality and quantity, and subsequently biodiversity (Campbell et al., 2017; Firbank et al., 2008). Landscape fragmentation has affected species habitats and biodiversity (Haddad et al., 2015; Hoang et al., 2023), and the global food system is responsible for one third of global anthropogenic greenhouse gas (GHG) emissions (Crippa et al., 2021). In several intensive agricultural areas, nitrogen and phosphorus losses pollute air, soil, and water, leading to biodiversity loss in terrestrial and aquatic ecosystems (Schulte-Uebbing et al., 2022; de Vries, 2021). Environmental degradation has subsequently resulted in a decrease of ecosystem services essential for maintaining life on Earth (IPBES, 2019; Mooney et al., 2009), a decline in human health through exposure to, or intake of, pollutants (Kampa & Castanas, 2008), a decrease in the potential of agricultural production (Arora, 2019), and with that the potential of feeding the world's growing population within safe ecological boundaries. Hence, there is a high need to mitigate agriculture-driven

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environmental pressures and restore degraded environmental systems while producing enough food for the world's population (Choi, 2004; Erisman, 2021).

Recognizing the agricultural and environmental challenges at European and global scales, European environmental policymaking is increasingly driven by long-term sustainability goals (EEA, 2017). In the last two decades, the European Union (EU) has applied a broad set of regulations and incentives to improve the agricultural sustainability (Eyhorn et al., 2019), including the National Emission Ceilings Directive, the Birds and Habitats Directive, the Water Framework Directive, Sustainable Use of Pesticides Directive, and the European Climate Law. While progress has been made in many respects, the coming 10–20 years will be most critical for making the transition to a global food system in which impacts of food production remain within planetary boundaries. Policymakers at all levels need to create clear, science-based, and harmonized regulatory frameworks, but also dynamic policies that incentivize innovation in technologies, practices, and business models (Dobermann et al., 2022).

While environmental policy is mainly focused on results and performance at the European level, policies that have direct implications for farmers tend to focus more on practice-based approaches. Building on the success of Key Performance Indicators (KPIs) in business organizations, various initiatives are exploring their potential use in agriculture, specifically at the farm level, to monitor or reward farmers' performance in achieving targets related to environmental quality (Payraudeau & van der Werf, 2005; Erisman et al., 2016; Ros et al., 2022).

This paper proposes a blueprint for the development and operationalization of a European KPI-based farm-level sustainability assessment and monitoring system. First, we present the rationalization for utilizing KPIs to implement performance-based agricultural policies. Then, by synthesizing findings from studies examining farm-level sustainability assessment tools, we introduce (1) eight guiding principles for developing and operationalizing a European agricultural KPI system built around the advantages of performance-based policies and (2) a 5-step process framework that incorporates the formulated principles to guide the system's development and operationalization. While we prioritize recent, high-quality peer-reviewed literature, we also incorporate relevant grey literature, recognizing its value in capturing emerging insights and advancements not yet published through formal peer review. Finally, we discuss the essential components of a policy structure supporting the proposed system.

While the proposed system focuses on the sustainability performance of farms and farming systems in the regional and national context, we acknowledge that many agricultural environmental impacts are driven by systemic factors beyond the farm, including supply chains, market dynamics, and consumer behavior. Addressing these broader pressures, drivers, and responses from all stakeholders in the food system is essential to achieving transformative change in the global food system. However, the farm (system) level remains a critical scale for evaluating and implementing sustainability interventions due to its direct connection to resource management practices (Del Prado et al., 2013). Farm-level assessment and monitoring tools that provide actionable insights are particularly relevant because they operate on the same scale at which management decisions are made. Additionally, regulation and monitoring at the farm level offer a transparent and decentralized approach that can aggregate into significant regional and global impacts when scaled effectively. This paper proposes an actionable and pragmatic system that complements broader systemic solutions and provides a foundation for more granular, performance-based policymaking.

2. Utilizing key performance indicators to implement performance-based agricultural policies

2.1. Current sustainability efforts in Europe

At the European-level, environmental policy is focused on results and

performance instead of compliance and rules. For example, each EU member state is mandated to determine their own agri-environmental strategy based on the overarching goals of the CAP (Carey, 2019) and establish their own biodiversity targets and conservation areas under the Birds and Habitats Directives (Bijlsma et al., 2018). However, this delegatory approach is not always applied in policies and regulations that govern the farm level. The agricultural policies that directly impact farmers are generally practice-based, i.e., they prescribe or reward the implementation of certain land management practices to achieve desired outcomes (Kleijn et al., 2020). These policies include the Nitrate Directive, which prescribes maximum manure application rates, and the Common Agricultural Policy's (CAP) eco-schemes, which provide financial support for farmers who observe sustainable agricultural practices.

At the farm-level, practice-based policies contrast with performance-based policies that reward or regulate farmers based on their environmental performance, which is measured relative to predetermined targets using farm-level performance indicators. In DPSIR terms (Driver-Pressure-State-Impact-Response framework; Kristensen, 2004), both practice- and performance-based policies are *responses* that aim to achieve improvements in environmental *state* (e.g., nitrate concentration in groundwater). The difference between the two is that practice-based policies aim to affect change in *drivers* (e.g., nitrogen fertilizer applied) whereas performance-based policies focus on the resulting *pressures* (e.g., nitrogen lost to the environment).

2.2. Potential advantages of performance-based policies

The literature on agri-environmental policy instruments has identified many potential advantages of performance-based policies compared to practice-based policies with similar objectives. These include:

- giving farmers the freedom to implement the sustainability measures they see fit based on their specific local expertise (Janus & Holzapfel, 2016; Herzon et al., 2018),
- providing motivation to farmers regarding sustainable farming practices and fostering innovation (Janus & Holzapfel, 2016; Burton et al., 2008; Herzon et al., 2018),
- integrally steering towards improvements necessary to reach all major sustainability targets and maintain ecologically functional landscapes instead of benchmarking farmers against themselves or their peers (Erisman et al., 2016; Kleijn et al., 2020),
- enabling an evidence-based learning-by-doing approach for pursuing multiple sustainability goals simultaneously (Kleijn et al., 2020),
- increasing the likelihood that payments will lead to environmental achievements (Payraudeau & van der Werf, 2005; de Olde et al., 2018; Herzon et al., 2018), and
- creating and strengthening links among different stakeholders influencing the sustainability of agricultural systems (Herzon et al., 2018).

Despite these potential benefits, the prevalence of practice-based agricultural policies at the farm-level persists due to the challenges involved in designing and implementing an appropriate performance verification system predicated on environmental monitoring at the farm or field level (Herzon et al., 2018; Jeffery & Verheijen, 2020). For example, sowing of cover crops can be prescribed to reduce erosion risk, and conducting compliance checks to verify whether the planting has been done is more feasible than measuring erosion directly in the field with specific instruments and techniques (Bélanger et al., 2012).

2.3. Moving towards performance-based policies with key performance indicators

In agriculture, key performance indicators (KPIs) are farm-level pressure indicators calculated using simple or complex models and

assessment methods (e.g., IPCC 2006 Guidelines) that estimate and aggregate impacts per environmental issue (e.g., CO₂ equivalent emissions) based on farm-level data regarding farming practices, inputs, yields, as well as landscape-level (geo-spatial) data such as proximity to water sources and protected areas (Payraudeau & van der Werf, 2005; Erisman et al., 2016; Ros et al., 2022). KPIs are often used in farm-level sustainability assessment tools developed by researchers and private value-chain stakeholders to help farmers understand, assess, and manage their farm's sustainability performance in relation to the governing authority's long-term sustainability goals (e.g. RISE, SAFA, Cool Farm Tool) (Kelly et al., 2018; Alexandropoulos et al., 2023; de Olde et al., 2018).

KPI methodologies infer farm sustainability performance using data already collected by agricultural stakeholders instead of relying solely on environmental monitoring at the farm- or field-level. For example, nitrate leaching to groundwater is too costly to measure at the farm scale but can be estimated as a field- or farm-level KPI given nitrogen input and output, crop type, soil type, groundwater depth, and soil compaction (Ros et al., 2020). The EU and its member states can use KPIs to cost-effectively monitor the sustainability performance of individual farms, set context-based targets, distribute financial rewards, compare farm performance to identify, adapt, and upscale better practices from other farmers, and evaluate and improve policy measures and instruments (Kelly et al., 2018).

2.4. Data limitations and the common European agricultural data space

One significant challenge currently facing farm-level sustainability assessment tools is the lack of data in the public data space that can be used to derive appropriate, meaningful, and relevant KPIs (Kelly et al., 2018; de Olde et al., 2018). As a result, farmers often need to provide extensive basic and technical data through surveys and questionnaires, which means that they cannot complete assessments without a considerable time investment or expert assistance (FarmAC, 2016; Häni et al., 2003; Zahm et al., 2008; FAO, 2014). These limitations are major obstacles for the widespread adoption of farm-level sustainability performance assessment tools and KPI methodologies (Alexandropoulos et al., 2023; de Olde et al., 2016b; de Olde et al., 2018).

The EU's efforts to digitalize the agricultural sector hold promise for overcoming data limitations. The EU is taking steps towards digitalizing European agriculture by bringing together high-tech technologies such as information and communication technologies (ICT), system integration, remote sensing, and big data (Șerbănel, 2021; Kelly et al., 2018). An important component of this digital transition is the development of a common European agricultural data space to enable data-sharing between different data systems and stakeholders in the agricultural sector (i.e., farm management systems and their providers, national/regional data hubs operated under public-private partnerships, business-to-business ecosystem platforms, and high-value datasets from the public sector) (European Commission, 2023). By consolidating real-time data from various sources (including inputs, yields, pesticide and antibiotic usage reported by farmers and supply-chain actors, routine laboratory data collected by governments, remote sensing-derived information such as land cover and crop identification, and geo-spatial data such as groundwater depth and proximity to protected natural areas), the common data space significantly enhances the availability and accessibility of data with which KPIs can be derived, facilitating the widespread adoption of KPI methodologies (Robling et al., 2023; European Commission, 2023; Kelly et al., 2018).

The development of the common European agricultural data space is ongoing and funded through the Digital Europe Program (DIGITAL) (European Commission, 2021). In 2023, the requirements for developing the data space were identified and preparatory actions were taken (e.g., proposals for conceptual approaches for the set-up of the data space, development of a multi-stakeholder governance scheme for the data space, and development of a roadmap for the deployment of the

data space) (European Commission, 2022b). Since May 2024, a consortium selected by the European Commission is working on the deployment of the data space, including building the required technical infrastructure and providing access to high-value datasets relevant to the agricultural domain (European Commission, 2024).

The advent of the common European agricultural data space paves the way for the evolution of European agricultural policy towards performance-based sustainability assessment and monitoring at the farm level. This evolution necessitates the development and operationalization of a comprehensive set of KPIs for assessing and rewarding farm-level sustainability. KPI development and operationalization processes are well-documented by the literature on farm-level sustainability assessment tools (Alexandropoulos et al., 2023; Binder et al., 2010). However, there is a need for a synthesis of this literature to develop guiding principles and a process framework for a large-scale KPI system that can effectively and efficiently meet a range of large-scale, long-term sustainability targets. Drawing upon the literature on farm-level sustainability assessment tools, this paper endeavors to guide this evolution by proposing a blueprint for the development and operationalization of a Europe-wide KPI-based farm-level sustainability assessment and monitoring system. The following section introduces and describes the proposed blueprint, including the eight guiding principles and the 5-step process framework.

3. Establishing a European agricultural KPI system: 8 guiding principles and 5 key steps

Given the complexity of sustainability and the number of sustainability assessment tools that exist, many different practices, processes, and methods have been used to conduct and deliver farm-level assessments. Review papers that compare existing tools have identified several important characteristics that can be used to differentiate between tools based on which practices, processes, and methods they employ. (Marchand et al., 2014; Binder et al., 2010; Alexandropoulos et al., 2023; de Olde et al., 2016b). Leveraging insights from these studies, we argue that adhering to certain principles regarding these characteristics is essential for developing an inclusive, integrated, and scalable European Agricultural KPI system. These principles can be found in Table 1 alongside the description of the characteristics and their purpose related to the potential benefits of performance-based policies.

To enhance guidance, it is important to delineate the steps of development and operationalization in which the eight guiding principles can be satisfied. Hence, drawing on the logic of governance and sustainability processes, such as the Plan-Do-Check-Act (PDCA) process and the adaptive management framework (ISO, 2015; Wiens et al., 2017), and synthesizing the common steps used in many farm-level sustainability assessment tools (Table 1), we present a 5-step framework that can guide Europe through the development and use of a KPI system for monitoring and rewarding farm-level sustainability (Fig. 1). We illustrate the steps of our framework alongside the steps of the PDCA process, as this logic will be familiar to governance and sustainability audiences. The structure of the following sections follows the steps within the framework. In each section, we describe the step, discuss the reasoning behind the guiding principles associated with the step, give relevant examples from farm-level sustainability assessment tools, and outline the potential roles of various stakeholders.

3.1. Step 1 – Establish sustainability goals

Step 1 of the framework, the first step in the planning phase (Fig. 1), involves defining the concept of sustainability, encompassing environmental, economic, and social dimensions, or a combination of the three (Binder et al., 2010). The Food and Agricultural Organization (FAO) and other international organizations stress that sustainability assessments of agricultural systems should account for the balance of environmental, social, and economic dimensions due to the interconnected nature of

Table 1

Eight guiding principles for developing and operationalizing the European Agricultural KPI system.

Tool characteristic and description	Guiding principles	Purpose of the guiding principle
Definition of sustainability <i>The concept of sustainability adopted (Marchand et al., 2014)</i>	A. Three-pillar (environmental, economic, social)	Allows for an integrated assessment and reveals interactions between sustainability indicators (trade-offs, synergies)
Goal setting <i>How were the goals set for the sustainability assessments? (Marchand et al., 2014)</i>	B. Top-down <i>Goals are predefined and derived from the definition of sustainability adopted (Binder et al., 2010)</i>	Ensures that the system contributes to EU- and national-level goals
Indicator interaction <i>Is interaction between indicators addressed in the tool? (Marchand et al., 2014)</i>	C. Decision support system (e.g., specific recommendations)	Supports farmers in their decision-making by identifying measures that work towards multiple sustainability goals (de Olde et al., 2018)
Stakeholder involvement <i>Are stakeholders involved in the development of the tool? (Alexandropoulos et al., 2023)</i>	D. Stakeholder involvement in multiple phases of development	Gives farmers agency and makes use of their knowledge base and specific local expertise
Applicability <i>Is the tool suitable for use in multiple countries and agricultural sectors? (de Olde et al., 2016b)</i>	E. Multinational and multisectoral	Ensures applicability to contiguous large areas, which is crucial for large-scale conservation and restoration
Reference values <i>The method used to determine reference values against which indicator scores are benchmarked (Acosta-Alba & van der Werf, 2011)</i>	F. Normative reference values <i>Reference values are derived from policy documents and scientific literature (Acosta-Alba & van der Werf, 2011)</i>	Benchmarks KPIs against improvements necessary to maintain ecologically functional landscapes (Acosta-Alba & van der Werf, 2011)
Target group <i>The groups of people/stakeholders who could be interested in using the tool (Alexandropoulos et al., 2023)</i>	G. Includes farmers	Ensures user-friendliness, which is crucial for the system's adoption and effectiveness (de Olde et al., 2016b)
Indicator relevance <i>The degree to which the tool's indicators increase the likelihood of accomplishing the goals which are implied by them (Hjørland and Christensen, 2002)</i>	H. Adaptive management	Adjusts KPI formulas and targets to improve indicator relevance over time

these factors and address several key issues such as food security and living wages (Bonisoli et al., 2018). Tools oriented towards environmental issues tend to have a poor emphasis on social and economic indicators such as labor and workload (Chopin et al., 2021). A comprehensive definition of sustainability will ensure that the KPI system's goals align with the EU's multifaceted vision for sustainable agriculture and that the provided assessment can serve as the basis on which farmers develop and implement their integrative measures. According to Silvestri et al. (2022), such an integrated approach offers an easier transition to sustainability compared to formulating goals in terms of a single sustainability issue and optimizing solutions solely for it.

Step 1 also involves using the chosen definition of sustainability to theoretically derive sustainability goals (Binder et al., 2010). This can be done using a predefined set of goals (top-down) or in a multi-stakeholder transdisciplinary process (bottom-up) (Binder et al., 2010). The advantage of a top-down approach for the European Agricultural KPI system is

that it allows farm-level targets to be linked to the predefined set of EU- and national-level sustainability performance targets. The bottom-up approach to goal setting, on the other hand, could lead to a lack of alignment and coherence with overarching EU and national sustainability targets, potentially resulting in fragmented or inconsistent objectives across different regions or stakeholders. This concern is particularly relevant when drivers and impacts are spatially disconnected, for example when excessive nutrient loadings in coastal areas are caused by inputs from alongside the flow path of water throughout the landscape. The predefined set of EU- and national-level environmental, social, and economic goals that should be adopted for the European Agricultural KPI system include the general and specific performance-based objectives set out in CAP legislative proposal, the Favorable Conservation Statuses (FCSs) established by member states under the Birds and Habitats Directives, etc. (see Table 2 for full list of EU-level sustainability performance targets). Adopting this set of (inter) national sustainability targets in Step 1 is vital for creating a scalable end-to-end system that is both ambitious and actionable. To ensure coherence and consistency, the tasks of defining sustainability and adopting a set of goals should be undertaken by the European Commission.

3.2. Step 2 – Explore linkages between goals, available data, and production practices

Following the establishment of sustainability goals, the next step involves exploring linkages between these goals, the farm-level and geospatial data available through the common European agricultural data space, and specific agricultural production practices. The relationships between sustainability goals and available data are the underlying foundation on which KPIs are developed in the next step of the framework (Step 3, Fig. 1). Identifying these relationships beforehand is crucial for ensuring that the KPIs developed are both relevant and grounded in real-world agricultural data. Hence, this step aims to pinpoint the most relevant data points for measuring progress towards the sustainability goals identified in Step 1. For this, the causal relationships between the goals and the data must be explored using the existing knowledge base regarding the functioning of ecosystems and the interaction between their various parts, including the interaction across scales, such as between local/regional ecosystems and global processes. An example of accounting for interactions across scales is linking the regional goal of achieving critical nitrogen deposition limits with data on fertilizer application rates (and other relevant data points) from farms in the surrounding area.

Step 2 also aims to lay the foundation for a decision support system by identifying relationships between specific agricultural production practices and the farm-level data available for KPI calculations. Farming practices often have multifaceted effects on multiple sustainability goals. Trade-offs between various sustainability issues that farmers frequently encounter, such as optimizing crop yields while minimizing environmental impacts, or enhancing biodiversity without sacrificing economic viability, make it challenging for farmers to decide on which sustainability measures to implement (de Olde et al., 2018; Binder et al., 2010). Decision support approaches play a pivotal role in navigating these trade-offs by providing farmers with specific, actionable recommendations tailored to their unique circumstances (de Olde et al., 2018; Alexandropoulos et al., 2023). These recommendations can be developed by consulting the existing knowledge base and identifying the effects of agricultural production practices on the calculated KPIs. The BBWP tool (*Bedrijfs Bodem Water Plan*) developed by Ros et al. (2020) uses this approach to offer insights into which practices will have the most positive effect on KPIs for groundwater quality, surface water quality, and groundwater recharge. For instance, the decision support system might suggest precision agriculture techniques to reduce chemical inputs while maintaining crop productivity or suggest different measures for increasing soil organic content based on the regional

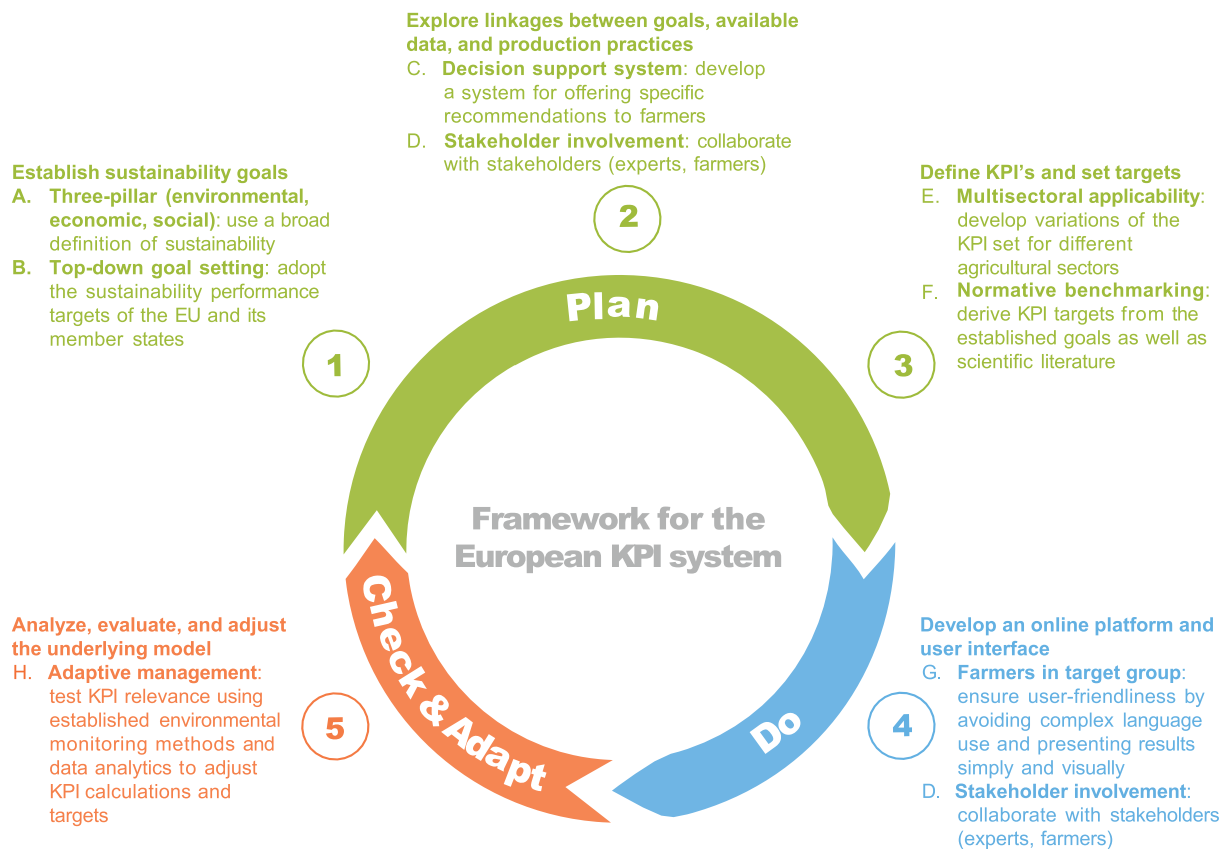


Fig. 1. Framework to guide the development and operationalization of the European Agricultural KPI system.

climate. Integrating decision support approaches into the KPI system can give farmers access to a holistic view of the potential consequences of their choices, enabling them to make informed decisions that balance competing sustainability goals and prioritize actions that yield the greatest overall improvement to sustainability.

The proposed linkage exploration processes must be carried out collaboratively with the participation of stakeholders from various sectors, including farmers, researchers, and policymakers. A collaborative and multi-level exploration process can leverage the specific knowledge base that farmers possess and act as a process of joint learning and knowledge development (de Olde et al., 2016a; Pretty, 2007). Identifying interactions collaboratively also increases the system's legitimacy and acceptance and can stimulate stakeholders to recognize and accept their role and responsibility for achieving a more sustainable practice (de Olde et al., 2016a, 2018). During this step, certain techniques can be used to facilitate collaboration, for example, the Delphi approach has been proven to help participants discuss, identify, and reach a consensus on causal interactions in food systems (Allen et al., 2018). Alternatively, Gari et al. (2015) found that the DPSIR framework can be used during stakeholder meetings to define variables in DPSIR terms and identify the causal interactions among them. Challenges in these participatory processes such as power asymmetries and complex group dynamics should not be underestimated and require process facilitation (Ansell & Gash, 2007).

3.3. Step 3 – Define KPIs and set targets

Step 3 is the final step of the planning phase wherein KPIs and their target-setting methodologies are defined. KPIs are defined by their calculations, which are weighted formulas that describe the causal relationships identified in Step 2. These formulas make use of relevant

farm-level and geo-spatial data points to aggregate impacts per sustainability issue (e.g., soil quality, farmer income) and estimate effects on sustainability goals (see Fig. 2). Common approaches to quantifying the causal relationships between available data and sustainability goals include process-based models (reliant on understanding and modeling the underlying processes or mechanisms that drive performance), empirical and data-driven models and methods (reliant on historical data, statistical analysis, and data mining techniques to identify meaningful performance indicators), and expert-judgment based methods (reliant on the knowledge, experience, and expertise of individuals or groups with relevant knowledge in the agricultural domain) (Ros et al., 2022). During this step, researchers and academics should have the role of providing these scientific methodologies for KPI calculations. Some example KPIs calculated using data sourced from farmers (farm management systems) and EU- (FADN, FES) or national-level (GO, CDKLW) databases can be found in Table 3 below. Ultimately, Step 3 should produce a set of KPIs that cover the range of sustainability goals adopted in Step 1.

Variations of this KPI set must be developed for different agricultural sectors to address the unique sustainability challenges of different sectors, such as crop diversity in arable farming versus feed production in livestock farming. For example, the KPI-K program in the Netherlands (*KPI's Circular Agriculture*) specifies the sectors (crop, dairy, pig, and poultry) to which each of its indicators is relevant (van Doorn et al., 2021). Similarly, the Biodiversity Monitor (*Biodiversiteitsmonitor* in Dutch) has two variations of its indicator set, one for crop production and one for dairy farming (van Laarhoven et al., 2018; Erisman et al., 2016). This approach ensures horizontal scalability by making the system suitable for use in multiple agricultural sectors and therefore applicable to contiguous large areas, which is crucial for ensuring large-scale conservation and restoration.

Table 2

Predefined set of EU-level sustainability performance targets, taken from [Erisman et al. \(2021\)](#) and CAP's monitoring and evaluation framework ([European Commission, 2022a](#)).

Theme	Policy	Performance target(s)
Soil quality	CAP	Soil organic carbon in arable land, soil erosion by water
Farmer income	CAP	Agricultural entrepreneurial income, agricultural factor income, rural GDP per capita, degree of rural poverty, rural employment rate, farm income variability
Biodiversity	CAP	Farmland bird index, share of grassland in total utilized agricultural area, share of ecological focus area in arable land, crop diversity
	Birds and Habitats Directives	Favorable Conservation Statuses (FCSs) established by member states for specific nature areas and species
Agricultural production	CAP	Total factor productivity in agriculture, value added for primary producers in the food chain
Water	Water Framework Directive/Nitrates Directive	Chemical (threshold for > 100 pollutants, excluding nutrients) and ecological status of surface water
	CAP	Condition of groundwater bodies (50 mg N/liter threshold) Balance between withdrawal and replenishment of groundwater
Climate	2030 Climate and Energy Framework CAP	Water abstraction in agriculture, water quality Reduce GHG emissions by at least 40% by 2030 compared to 1990 levels Net greenhouse gas emissions from agricultural soils
Circularity	Circular Economy Action Plan	Promote the circular economy in the EU, focusing on waste reduction, recycling, and sustainable resource management

Step 3 also involves determining target values for the KPIs. [Acosta-Alba & van der Werf's \(2011\)](#) review of farm-level sustainability assessment methods uncovers the variety of target setting methods used. One method, *relative target values*, benchmarks farmers with themselves or their peers (e.g., 30% less fertilizer or 80% less pesticides) without assessing whether the measures will add up sufficiently on an ecosystem/landscape scale ([Acosta-Alba & van der Werf, 2011](#); [Kleijn et al., 2020](#)). Thus, even if every farmer reaches their target, it is not guaranteed that this improvement is enough to restore biodiversity and ecosystem functions at a large scale ([Kleijn et al., 2020](#)). For example, despite a 60% reduction of ammonia emissions from agricultural sources in the Netherlands since 1990, the critical loads in nature areas are still exceeded and surface and groundwater nitrogen targets are still out of reach ([Erisman, 2021](#)). This illustrates that relative targets may lead to inefficient use of sustainability efforts by rewarding incremental improvements without addressing the underlying regional and local environmental issues.

In contrast, *normative target values* allow the assessment of a single system with previously defined values derived from scientific literature and policy objectives ([Acosta-Alba & van der Werf, 2011](#)). Normative target values are based on scientific knowledge regarding what is needed from an ecosystem perspective. In the case of the proposed KPI system, normative KPI targets can be derived by translating the environmental goals adopted in Step 1 into farm-level targets. This ensures that if all farmers achieve their KPIs, agriculture-driven environmental degradation would be effectively halted, allowing ecosystems to function within safe ecological boundaries. One example is the regionally defined boundaries for critical N inputs in view of environmental impacts on nature, groundwater and surface water ([de Vries, 2021](#)). By using normative targets, the KPI system can overcome the limitations of relative targets and ensure that implemented measures result in improvements necessary to maintain ecologically functional landscapes and reach the EU- and national-level sustainability goals ([Kleijn et al., 2020](#)).

To effectively implement normative target values, it is crucial to use spatially explicit methods that calibrate the target values of each farm

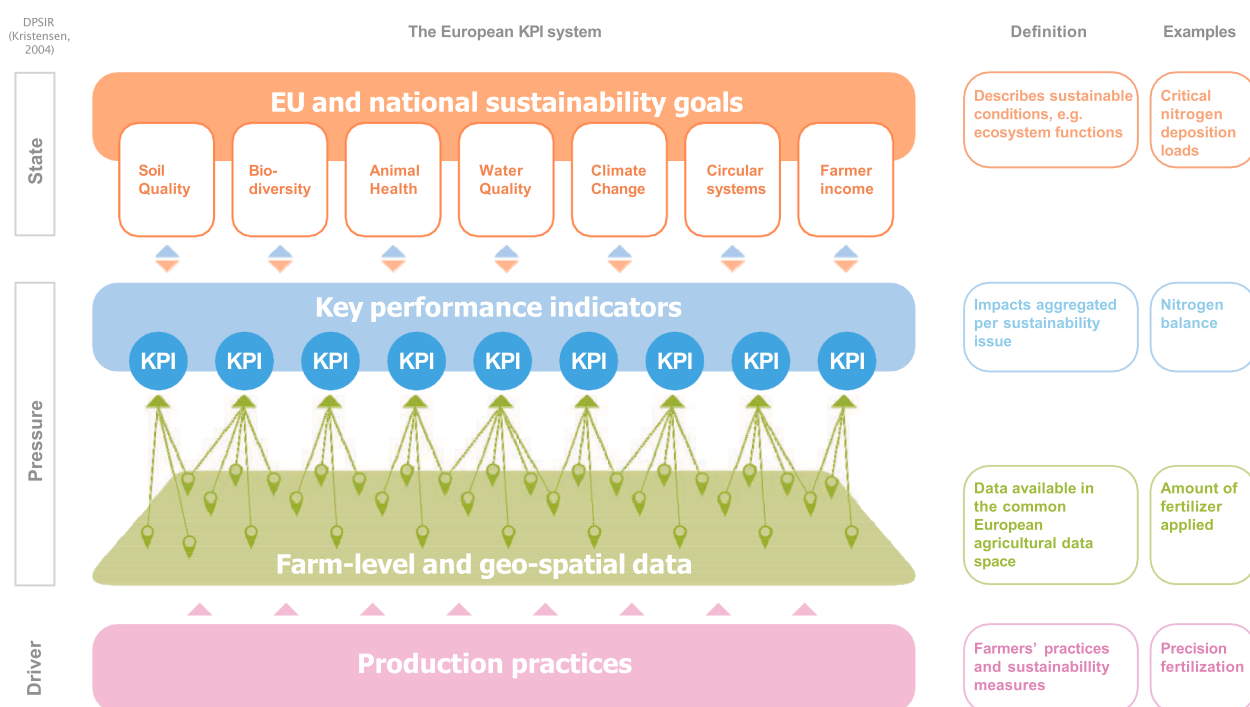
**Fig. 2.** The proposed European Agricultural KPI system design.

Table 3

Example farm-level sustainability KPIs. Compiled from Robling et al.'s (2023) review of farm-level sustainability indicators and the operational KPI sets introduced by van Doorn et al. (2021) and van Laarhoven et al. (2018).

Theme	KPI	Units/ factor measured	Data source (s)	Variables
Soil quality	Soil organic matter (Dolman et al., 2014)	g/100 g soil	FMS ¹ , GO, CDKLW	Green manure, organic fertilizer, crop residues
Farmer income	Gross output + subsidies (Dillon et al., 2010)	Euro/farm	FADN/ FES	Output, subsidies, costs, wages (Euro)
Biodiversity, Climate, Water, Circularity	Nutrient (Phosphorus (P), Nitrogen (N), Potassium (K)) balance/surplus per hectare farmed area (van Doorn et al., 2021)	kg N, P, K input/Kg output	FMS, GO, CDKLW	N, P, K surplus, kg/ha
Biodiversity	Yearly Plant Protection Products (PPP) use (Burman et al., 2021)	kg/ha	FADN/ FES ²	PPP inputs, Euro
Agricultural production	Output value – total costs per hectare (Ehrmann 2008)	Euro/ha	FADN	Output, costs, agricultural land (Euro, ha)
Water	Yearly irrigation/water use (Marton et al., 2016)	liter (L)/ha	FES	Irrigated area, ha
Climate	Permanent to temporary grass area (Barnes & Thomson 2014)	ha, %	FADN, FES	Permanent pasture and meadow/ temporary grass, ha
	Non-renewable energy use (Dolman et al., 2014)	% of total energy	FADN/ FES	Total fuel purchases, Euro
Circularity	Percentage of protein from your own country/area (van Laarhoven et al., 2018)	kg/ha/ year, %	CDKLW	Feed produced on farm, feed consumption

¹ FMS – Farm management systems, FADN – Accountancy Data Network, FES – Farm Economic Survey, CDKLW – Central Database KringloopWijzer, GO – Gecombineerde Opgave.

² Data sources in bold indicate that the data source is temporary and/or requires additional data to measure the indicator.

according to the environmental challenges and goals of its region. For example, nitrogen emission indicator targets should be adjusted depending on the proximity of the region to biodiversity hotspots, while soil organic content targets should be adjusted based on the regional climate (Voltr et al., 2021). Hence, an important part of Step 3 includes analyzing the literature to find the most suitable and scientifically sound methodologies for differentiating KPI targets across European landscapes and regions. Researchers and academics should conduct these literature reviews and identify the most suitable target differentiation methodologies. Some examples from sustainability assessment tools currently available to farmers include: (1) the Ecological Scarcity Method (ESM), which supports spatial differentiation of target values through the implementation of six region-specific scarcity categories based on Swiss freshwater consumption targets (Acosta-Alba & van der Werf, 2011), (2) the Biodiversity Monitor's method for distinguishing between types of soil (peat, clay, and sand) when assigning targets to farms (van Laarhoven et al. 2018), (3) RISE and FAO's SAFA methods for

adjusting KPI scores based on region-specific environmental and economic conditions like water scarcity and farmer income (de Olde et al., 2016b; Scialabba et al., 2013), and finally, (4) Erisman et al.'s (2021) inverse modeling system for calculating ammonia emission targets per region based on proximity to protected Natura 2000 areas and the nitrogen deposition targets for these areas. When spatially explicit targets cannot be derived due to knowledge gaps, relative targets can be used.

3.4. Step 4 – Develop an online platform and user interface

In the implementation phase (Step 4, Fig. 1), the theoretical work done during the planning phase is translated into a functional information system (IS) product. This IS product, preferably an online platform, serves as the user-facing component of the system architecture to maximize accessibility and usability. This platform's development should follow a structured systems engineering approach, encompassing requirements analysis, system design, implementation, testing, and validation.

The backend architecture should integrate with the common European agricultural data space via optimized database queries, enabling efficient data ingestion (Fig. 3). Embedded algorithms in the platform's processing layer will compute KPI scores and generate farm-specific decision support outputs. The interface layer, designed with human-centered design principles, must align with the usability requirements of the target users, i.e., farmers.

According to de Olde et al. (2016b), user adoption is influenced by the perceived relevance and intuitiveness of the interface. To enhance usability, the user interface (UI) should present outputs using a distance-to-target visualization format using spider, lollipop, gauges or other appropriate visualisations (e.g. Fig. 4) to intuitively display farmers' progress toward sustainability goals. Automated notifications can alert farmers when targets are achieved and allow them to easily and quickly claim the associated subsidies through the platform. To ensure the platform meets operational and user-centric requirements, iterative testing and validation with end-users should be conducted, incorporating feedback to refine functionality and usability.

3.5. Step 5 – Analyze, evaluate, and adjust the underlying model

It is vital that sustainability indicators and KPI systems adapt over time to take account of emerging issues and changing circumstances (Addison et al., 2020). Information collected through the online platform can be used to adapt and improve the system's effectiveness. Step 5 in the framework involves reviewing progress to date and adapting the KPIs, data collection methods, and sometimes even the sustainability goals when necessary. To this end, it is essential to not only measure changes in KPI scores, but also to monitor the actual reversing of environmental degradation over time. Environmental monitoring programs established in Europe, such as the Pan-European Common Bird Monitoring Scheme (PECBMS), can be used to measure those components of the environmental system for which the goals have been set in Step 1. This data on environmental quality can be coupled with the data on KPI scores collected through the online platform to test indicator relevance by checking whether advancements in KPIs in a region indeed lead to the desired changes in environmental state indicators of that region. This validation of environmental relevance can be performed by data analysts and researchers using data-driven statistical tests to determine how the KPIs (independent variables) influence the state indicators (dependent variables). When data analytics results show no improvement, or worsening, of environmental quality, the model underlying the system (Step 2) and therefore the KPIs and their target values (Step 3) should be adjusted accordingly to improve indicator relevance. This feedback loop is key for accommodating the uncertainties inherent in such scientific models. The European Commission should facilitate this adaptive management approach to enable stakeholders to regularly reassess and refine the underlying model considering emerging knowledge, shifting

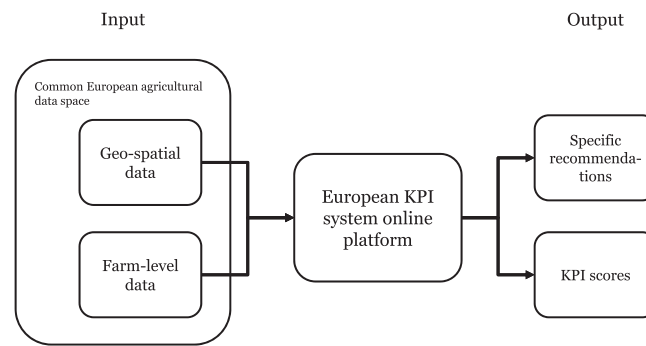


Fig. 3. Backend of the European Agricultural KPI system's online platform.

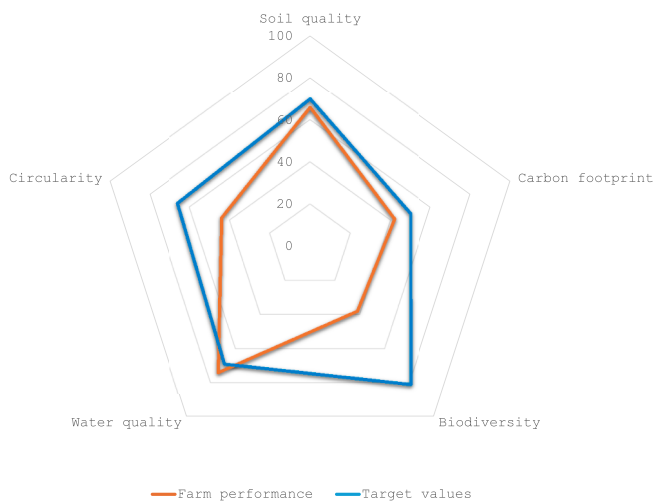


Fig. 4. Example distance-to-target KPI score visualization.

priorities, and unforeseen challenges, thereby enhancing its accuracy and effectiveness in guiding sustainable agricultural development.

3.6. Example from the Dutch dairy sector – The Biodiversity Monitor

An example of a KPI system similar to the one proposed in this paper is the Biodiversity Monitor, a set of KPIs developed to assess the sustainability of dairy farms in the Netherlands (Erisman et al. 2016; Kleijn et al., 2020). While the Biodiversity Monitor's development did not fully adhere to the guidelines outlined in this paper, it incorporates key elements such as normative reference values and top-down goal setting. For instance, the goals of the Biodiversity Monitor are adopted from the biodiversity targets set by the EU's Birds and Habitats Directive.

The Biodiversity Monitor features 17 specific farm-level KPIs, including nutrient losses, greenhouse gas emissions, and the occurrence and number of biodiversity-enhancing landscape elements such as trees and ditches. These KPIs were developed using farm specific and spatially explicit data from established datasets. Targets for the KPIs were determined normatively through extensive research to identify the values required to meet the adopted biodiversity goals.

The Biodiversity Monitor also demonstrates how farm-level sustainability assessments can translate into actionable incentives for farmers. For example, Rabobank, the leading agricultural bank in the Netherlands, offers interest rate discounts on loans to farmers who score

highly on Biodiversity Monitor KPIs, thereby promoting the adoption of sustainable management practices (Kleijn et al., 2020). Similarly, FrieslandCampina, the country's largest dairy company, rewards these farmers with higher milk prices (van Laarhoven et al., 2018).

In practice, farmers participating in the Biodiversity Monitor program undergo regular assessments based on farm-specific data, with KPIs measured through a combination of data submitted by farmers and on-site evaluations. The scoring system enables farmers to track their progress over time and adjust their management practices accordingly. In addition, the Biodiversity Monitor's indicators for biodiversity-enhancing landscape elements have been adopted to the KPI set developed in the KPI-K program (*KPI's Circular Agriculture*) funded by the Dutch Ministry of Agriculture, Fisheries, Food Security and Nature for use in future agricultural policy.

4. A supportive policy structure

The proliferation of agricultural data and the convergence of advanced technologies in Europe present opportunities for the widespread implementation of promising farm-level performance-based policies. However, the proposed KPI system requires more than just technological infrastructure; it necessitates a comprehensive policy structure encompassing interoperability standards, privacy regulations, financial incentives for participation, pilot programs, international collaboration, and synergy with existing initiatives. Here, we aim to explain the multifaceted requirements essential for developing a supportive policy foundation that encourages data sharing, digitalization, and the adoption of the proposed KPI system.

First and foremost, the policy structure would need to define clear objectives for the KPI system and establish a robust governance structure. Objectives would likely focus on promoting innovation, enhancing agricultural productivity and sustainability, improving market transparency, and empowering farmers with access to valuable data-driven insights. The governance structure would entail delineating the roles and responsibilities of various stakeholders, such as government agencies, industry associations, research institutions, and data platform developers and operators to ensure accountability and effective coordination.

Interoperability standards: Defining and implementing interoperability standards is imperative to ensure seamless data exchange and compatibility between different data sources and systems, including farm management software, supply-chain data platforms, and high-value datasets from the public sector. The ongoing efforts to develop the common data space need to be supported by policies that standardize how data is formatted and stored across European countries to

enable coupling of the datasets on certain data points like farm location. This will enhance the efficiency of the common data space and enable effective utilization of the integrated data for sustainability assessment.

Privacy regulations: Linking subsidy systems to performance indicators introduces ethical and democratic dilemmas, particularly if farmers are financially dependent on these subsidies to remain viable. While participation in a KPI system may be technically voluntary, the financial structure could lead to a situation of indirect coercion, where farmers feel compelled to share their data to avoid losing essential income. A potential solution to this issue would be to include safeguards that prevent undue financial pressure. These could involve ensuring that baseline subsidies are available independent of KPI performance, allowing farmers to meet basic operational needs while providing additional rewards for achieving sustainability targets. However, this approach may reduce the effectiveness of the KPI system by allowing large polluters to remain economically viable without fully addressing their environmental impact. Alternatively, farm-level data could be categorized as part of environmental data, which is publicly accessible in Europe in accordance with the Public Access to Environmental Information Directive (Bulens et al., 2019). This approach would ensure that critical data needed for sustainability assessments and policy-making is available without placing the burden of decision-making solely on individual farmers. However, this solution may not be preferable for farmers due to concerns about privacy and the potential misuse of their data. If such an approach is adopted, it would require stringent privacy regulations to protect farmers' interests, including anonymization of sensitive data and clear limits on how publicly accessible farm data can be used.

Financial incentives: Creating incentives for participation in the KPI system, such as financial support and subsidies for farmers, is crucial to encourage farmers, agribusinesses, and other stakeholders to contribute data to the common agricultural data space and leverage its benefits. However, the current European agricultural policy states that financial compensation (through CAP Pillars 1 and 2, i.e., direct income support and eco-schemes) should be based on costs and income loss incurred by the farmer stemming from the implementation of sustainable agricultural practices (Official Journal of the European Union, 2021). This form of payment is not compatible with the performance-based reward system proposed in this paper. Hence, the policy foundation supporting the KPI system must allow financial support based on the public goods (i.e., ecosystem services) provided by farmers who reach their KPI targets (Official Journal of the European Union, 2021).

Robust and fraud-resistant verification system: A data verification system must be specified within the policy structure to minimize KPI result manipulation while still decreasing administrative burdens and costs. The proposed KPI system relies on potentially fraud-prone data sourced from farm management systems and software. Implementing independent verification mechanisms can enhance the credibility and reliability of the KPI system. Independent auditors or verification agencies should be tasked with conducting audits and reviews, particularly upon reaching KPI targets, to verify the accuracy and reliability of reported data. These verification bodies should operate independently from the entities responsible for data collection and reporting to ensure impartiality and objectivity. Involving well-established eco-label certification bodies like Rainforest Alliance, Fairtrade, and Blue Angel could be advantageous given their track record in large-scale auditing and their status as non-governmental organizations.

Pilot programs: Pilot programs to test and improve the KPI system must be conducted before full-scale implementation. These programs serve as experimental testbeds, allowing stakeholders to assess the feasibility, effectiveness, and scalability of the KPI system in real-world agricultural settings. By piloting the system in diverse agricultural contexts, stakeholders can validate the relevance and applicability of KPIs across different agricultural sectors, regions, and farming systems. Furthermore, pilot programs provide an opportunity to test the feasibility and practicality of data collection methodologies, such as the one

proposed for measuring soil carbon by Fujisaki et al. (2023), and data management processes required to operationalize KPIs within the common agricultural data space.

Alignment with existing initiatives: European agriculture is already dotted with numerous initiatives, programs, and projects aimed at promoting sustainability, innovation, and digitalization (Horizon Europe, European Innovation Partnerships, European Network of Digital Innovation Hubs, etc.). Aligning these initiatives in a policy structure helps avoid duplication of efforts and resource fragmentation. This approach leverages the wealth of expertise they represent, drawing from successful methods and building on established infrastructures, networks, and methodologies. As a result, the KPI system benefits from the collective insights and lessons learned from both past and current efforts.

Supporting the European Agricultural KPI system with a robust policy structure is essential for ensuring its effectiveness, legitimacy, and long-term sustainability. By formalizing the KPI system in law, policymakers can demonstrate a firm commitment to innovative performance-based agricultural policy and arrange the allocation of resources and funding to support the implementation and operation of the KPI system. A policy structure that covers the points outlined above provides clarity and certainty to stakeholders, including farmers, policymakers, researchers, and industry representatives, regarding their rights, responsibilities, and obligations within the KPI system. It establishes clear rules and procedures for data collection, reporting, and verification, ensuring transparency, accountability, and fairness in sustainability assessment and monitoring. This policy structure would serve as the cornerstone for the development and operation of the common European agricultural data space and the KPI system, providing the legal and regulatory infrastructure necessary for their successful implementation.

5. Conclusion

The transition toward KPI-based farm-level sustainability assessment and monitoring presents both significant opportunities and challenges. The transition may face challenges and resistance from entrenched systems and legislation like the financial support structure of the CAP. Moreover, linking subsidies to KPI targets rather than CAP's direct income support could risk making some farms economically nonviable, particularly those unable to meet sustainability goals in the short term. This underscores the need for a transitional period during which subsidies are gradually restructured, allowing farmers time to adapt their practices. Furthermore, exceptions may need to be considered for regions where food security is closely tied to national agricultural production, ensuring that the shift to performance-based policies does not compromise access to affordable food.

Farmer support and acceptance of a KPI system is another critical challenge. While stakeholder engagement is a guiding principle in the development of the proposed system, there is a risk that farmers may resist adoption if they perceive the financial burdens to outweigh the benefits. Ensuring trust, fairness, and transparency in the implementation process will be essential to gaining their support and securing broad participation.

Despite these challenges, the proposed system blueprinted here offers substantial opportunities, particularly in leveraging advancements in technology. Emerging tools, such as remote sensing and satellite imagery, provide new possibilities for collecting farm-level data efficiently and accurately. For instance, nature and landscape elements, such as ponds, hedgerows, and trees, can be automatically mapped and monitored using these technologies, reducing the administrative burden on farmers and improving the precision of sustainability assessments. Such innovations can enhance the system's effectiveness while lowering implementation costs.

The implementation of the proposed system will inevitably have managerial implications. These might include the emergence of farm-level sustainability consultants whose expertise farmers might

increasingly rely on to interpret KPI data and optimize practices, or a labor skill shift caused by the system favoring tech-savvy farm managers capable of navigating digital platforms and data analytics and sidelining traditional farming expertise. The system might enable real-time policy adjustments whereby policymakers tweak incentives or regulations dynamically based on real-time KPI trends, which might reduce bureaucratic delays but also introduce uncertainty for farmers. Finally, farms with high KPI scores could leverage their sustainability performance for premium pricing or eco-labeling, creating a new competitive landscape.

In summary, while transitioning from current policies to a KPI-based system for farm-level sustainability monitoring may face resistance and economic challenges, its potential to drive meaningful improvements in sustainability is significant. To achieve success, policymakers must address the socioeconomic implications for farmers by ensuring a gradual and fair transition, accompanied by robust technological solutions that reduce barriers to participation. Further studies focusing on developing robust and relevant KPIs using the data points provided by data-sharing initiatives like the common European agricultural data space are warranted to enhance the feasibility of performance-based sustainability assessment and monitoring at the farm level.

CRedit authorship contribution statement

B.T. Barış: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **S.M. Marselis:** Writing – review & editing, Writing – original draft, Supervision. **G.H. Ros:** Writing – review & editing, Supervision, Resources. **A. van Doorn:** Writing – review & editing, Resources. **J.W. Erisman:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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